

Soil Sorption Estimator (SSE)

Requirements:

- SSE is a Streamlit-based web predictive platform that only requires a web-browser.
- Requires **internet connection** for deployment or online hosting.

GUI of the SSE Prediction Platform:

The screenshot displays the user interface of the Soil Sorption Estimator (SSE). On the left, there is a sidebar with a logo showing chemical structures and soil, titled 'LogK_{oc} Soil Sorption Estimator (SSE)'. Below the logo, it says 'Select Regression Model' with a dropdown menu currently set to 'PLS-QSPR Model'. An 'About' section lists model details: 'PLS QSPR Model: Uses MolLogP, RingCount, fr_nitro_arom, MaxAbsEStateIndex' and 'q-RASPR Model: Uses RA function(LK)'. The main area is titled 'Soil Sorption Estimator (SSE)' and includes a subtitle: '*Predict LogK_{oc} using PLS QSPR or Univariate q-RASPR models from SMILES.' Under 'Input Method', there are two radio buttons: 'Single SMILES' (selected) and 'Upload Excel/CSV'. Below this is a text input field labeled 'Enter SMILES string:' and a 'Predict LogK_{oc}' button. At the bottom, there are links for 'Contact Info' and 'User Manual'.

Figure 1: Soil Sorption Estimator GUI

***Note:-** The user needs a valid SMILES string for the compound that is to be predicted.

Prediction Procedure:

1. To predict a single compound, the user can simply use **Single SMILES** as **Input Method** enter the SMILES string of the desired compound as input in the “Enter SMILES string” tab and then click the “Predict LogK_{oc}” button.
2. To predict a set of compounds, the user should prepare an Excel file (.xlsx format) containing the SMILES for each chemical in the first column of the first sheet with a proper heading. Then, the user must select “**Upload Excel**” as the input method and select the prepared file as input in the “Upload” tab by clicking on the button shown in **Figure 2**. Afterward, click on the “Predict LogK_{oc}” button to obtain the predictions.

Figure 2: Steps to Upload SMILES file (.xlsx) for prediction

Input File:

- Must be in .xlsx format.
- A suitable heading in the first cell of the first column (**A1**).
- Valid SMILES in the first column from **Row 2**.

Table 1: Sample input file for the compounds needed to be predicted

	A	B	C
1	SMILES		
2	CCCCC		
3	C(F)CCNC		
.....			
.....			

*The user can also refer to the “Sample.xlsx” file in the ‘data’ folder to prepare their input file.

View and Download Result:

Once the prediction is complete, the results can be viewed directly within the interface. The output will include details such as SMILES (for the input compounds), predicted LogK_{OC} values, and Applicability Domain (AD) status. Users also have the option to download the results in .csv format for single/batch process, which can be used for further analysis, record-keeping, or reporting purposes.

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Tool developed by



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